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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Third Semester B.Tech Degree Examination December 2020 (2019 Scheme)

Course Code: MAT201

Course Name: Partial Differential equations and Complex analysis

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions. Each question carries 3 marks*

Marks

- 1 Derive a partial differential equation from the relation $z = (x + y) f(x^2 - y^2)$ (3)
- 2 Solve using direct integration $\frac{\partial^2 u}{\partial x \partial t} = e^{-t} \cos x$ (3)
- 3 Solve $2z = xp + yq$. (3)
- 4 Write any three assumptions in deriving one dimensional heat equation. (3)
- 5 Show that an analytic function $f(z) = u + iv$ is constant if its real part is constant. (3)
- 6 Show that the function $u = \sin x \cos hy$ is harmonic. (3)
- 7 Find the Maclaurin series of $f(z) = \sin z$ (3)
- 8 Evaluate $\oint_C \ln z \, dz$, where C is the unit circle $|z| = 1$. (3)
- 9 Find all singular points and residue of the function $\operatorname{cosec} z$ (3)
- 10 Determine the location and order of zeros of the function $\sin^4\left(\frac{z}{2}\right)$ (3)

PART B*Answer any one full question from each module. Each question carries 14 marks***Module 1**

- 11 (a) Form the Partial differential equation by eliminating the arbitrary constants from $(x - a)^2 + (y - b)^2 = z^2 \cot^2 \alpha$ (5)
- (b) Solve $2xz - p x^2 - 2qxy + pq = 0$ (9)
- 12 (a) Solve $\frac{\partial^3 z}{\partial^2 x \partial y} = \cos(2x + 3y)$ (7)
- (b) Solve $x^2(y - z)p + y^2(z - x)q = z^2(x - y)$ (7)

Module 2

- 13 (a) Derive the solution of the one dimensional wave equation $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$ using variable separable method. (6)
- (b) An insulated rod of length l has its ends A and B maintained at 0°C and

100° C respectively until steady state conditions prevail. If B is suddenly reduced to 0° C and maintained at 0° C, find the temperature at a distance x from A at time t . (8)

- 14 (a) Derive the one dimensional heat flow equation. (6)
- (b) A tightly stretched string of length l with fixed ends is initially in equilibrium position. If it is set vibrating by giving each points a velocity $v_0 \sin^3(\frac{\pi x}{l})$. Find the displacement $y(x, t)$. (8)

Module 3

- 15 (a) Find an analytic function whose real part is $u = \sin x \cosh y$ (7)
- (b) Find the image of the strip $\frac{1}{2} \leq x \leq 1$ under the transformation $w = z^2$ (7)
- 16 (a) Check whether $w = \log z$ is analytic. (8)
- (b) Show that under the transformation $w = \frac{1}{z}$, the circle $x^2 + y^2 - 6x = 0$ is transformed into a straight line in the W plane. (6)

Module 4

- 17 (a) Integrate counter clockwise around the unit circle $\oint_C \frac{\sin 2z}{z^4} dz$ (7)
- (b) Find the Taylor series of $\frac{1}{1+z}$ about the centre $z_0 = i$ (7)
- 18 (a) Evaluate $\int_0^{1+i} (x - y + ix^2) dz$ along the parabola $y = x^2$. (7)
- (b) Evaluate $\oint_C \frac{\log z}{(z-4)^2} dz$ counter clockwise around the circle $|z - 3| = 2$. (7)

Module 5

- 19 (a) Find the Laurent's series expansion of $\frac{z^2 - 1}{z^2 - 5z + 6}$ about $z = 0$ in the region $2 < |z| < 3$ (5)
- (b) Evaluate $\int_0^{2\pi} \frac{d\theta}{\sqrt{2} - \cos \theta}$. (9)
- 20 (a) Evaluate $\oint_C \frac{z-23}{z^2 - 4z - 5} dz$ where $C : |z - 2 - i| = 3.2$ using Residue theorem. (5)
- (b) Evaluate $\int_0^\infty \frac{(x^2 + 2) dx}{(x^2 + 1)(x^2 + 4)}$. (9)



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Scheme for Valuation/Answer Key

Scheme of evaluation (marks in brackets) and answers of problems/key
Third Semester B.Tech Degree Examination December 2020 (2019 scheme)

Course Code: MAT201

Course Name: Partial Differential Equations and Complex analysis

Duration: 3 Hours

PART A

Answer all questions. Each question carries 3 marks

Marks

1 $z = (x + y) f(x^2 - y^2).$ (3)

Finding p and q (1).

Eliminating $f(x^2 - y^2)$ (1)

Forming the PDE ($py + qx = z$) (1)

2 $\frac{\partial^2 u}{\partial x \partial t} = e^{-t} \cos x.$ (3)

Integrating partially with respect to x

$$\frac{\partial u}{\partial t} = e^{-t} \sin x + f(t) \quad (1).$$

Integrating partially with respect to t

$$u = -e^{-t} \sin x + \int f(t) dt + g(x). \quad (2)$$

3 Forming the subsidiary equations (3)

$$\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{2z} \quad (1).$$

Solving the subsidiary equations and finding two sets of solutions

(1)

$$\frac{x}{y} = a$$

$$\frac{y^2}{z} = b.$$

Writing the solution of the equation $2z = xp + yq$ as

(1)

$$f\left(\frac{x}{y}, \frac{y^2}{z}\right)$$



(1).

(Marks shall be given to any alternate solution)

4 Assumption 1

(1)

(3)

Assumption 2

(1)

Assumption 3

(1)

5 u is constant, then $\frac{\partial u}{\partial x} = 0$, $\frac{\partial u}{\partial y} = 0$ (3)

(1).

By CR equations $\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} = 0$ and $\frac{\partial v}{\partial y} = \frac{\partial u}{\partial x} = 0$.

(1).

Therefore v is a constant. Hence $f(z)$ is a constant.

(1)

6 $u = \sin x \cosh y$. (3)

$$\frac{\partial u}{\partial x} = \cos x \cosh y.$$

$$\frac{\partial^2 u}{\partial x^2} = -\sin x \cosh y. \quad (1)$$

$$\frac{\partial u}{\partial y} = \sin x \sinh y.$$

$$\frac{\partial^2 u}{\partial y^2} = \sin x \cosh y. \quad (1)$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$

 U satisfies Laplace equation and hence harmonic. (1)7 $f(z) = \sin z$. (3)Finding $f^1(z)$, $f^{11}(z)$, $f^{111}(z)$, ..Finding $f^1(0)$, $f^{11}(0)$, $f^{111}(0)$, .. (1)

Substituting in

$$f(z) = f(0) + \frac{z}{1!} f^1(0) + \frac{z^2}{2!} f^{11}(0) + \frac{z^3}{3!} f^{111}(0) + \dots \quad (1)$$

$$\sin z = \frac{z}{1!} - \frac{z^3}{3!} + \dots \quad (1)$$

8 $|z|=1$, $z = e^{i\theta}$ $dz = ie^{i\theta} d\theta$ (1) (3)

Evaluating the integral using the above substitution



$$\oint_C \ln z \, dz = 2\pi i \quad (2)$$

$$9 \quad f(z) = \operatorname{cosec} z = \frac{1}{\sin z} \quad (3)$$

$$= \frac{1}{z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots} \quad (1)$$

$$= \frac{1}{z} \left(1 - \left(\frac{z^2}{3!} + \frac{z^4}{5!} - \dots \right) \right)^{-1}$$

$$= \frac{1}{z} + \frac{z}{3!} - \dots \quad (1)$$

$z=0$ is a pole.

Residue at $z=0$ is 1. (1)

$$10 \quad f(z) = \sin^4 \left(\frac{z}{2} \right) \quad (3)$$

$$f(z) = 0 \text{ for } z = 0, \pm 2\pi, \pm 4\pi, \dots \quad (1)$$

$$f'(z), f''(z), f'''(z) = 0 \text{ and } f^{(4)}(z) \neq 0 \text{ at these points.} \quad (1)$$

So the zeros are of the fourth order. (1)

(3)

PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

$$11(a) \quad (x-a)^2 + (y-b)^2 = z^2 \cot^2 \alpha \quad (5)$$

Finding p and q (3).

$$p^2 + q^2 = \tan^2 \alpha \quad (2)$$

$$(b) \quad f(x,y,z,p,q) = 2xz - p x^2 - 2qxy + pq \quad (1)$$

writing Charpits subsidiary equations (2) (9)

finding p and q from the subsidiary equations (3).

Substituting p and q in $dz = pdx + qdy$ (2).

Solving and finding $z = ay + b(x^2 - a)$ (1)

12(a) Integrating Partially with respect to x (2 times) , Integrating partially with respect to y (7)

Writing the solution . (3)

$$Z = \frac{-\sin(2x+3y)}{12} + x v(y) + u(y) + w(x). \quad (3)$$



Where $v(y)$, $u(y)$ are functions of y only
and $w(x)$ is a function of x only. (1)

(b)

$$x^2 (y - z)p + y^2 (z - x)q = z^2 (x - y).$$

Writing the subsidiary equations (1)

Forming two sets of solutions (4).

Writing the solution

$$xyz = f\left(\frac{1}{x} + \frac{1}{y}, + \frac{1}{z}\right) \quad (2).$$

Any other alternate solution will gets full marks.

Module 2

13(a) Derivation (6)

(b) The equation of Heat conduction $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ (1)

Initial condition $u(x,0) = \frac{100x}{l}$ (1) (8)

Boundary conditions

$$u(0,t) = 0 \text{ for all } t \quad u(l,t) = 0 \text{ for all } t \quad (1)$$

$$\text{Solution } u(x,t) = (c_1 \cos px + c_2 \sin px) e^{-c^2 p^2 t}. \quad (1)$$

Applying the conditions and writing the solution

$$U(x,t) = \frac{200}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin \frac{n\pi x}{l} e^{-\left(\frac{n\pi c}{l}\right)^2 t} \quad (4)$$

**** Give marks for any fair attempt irrespective of l (small L) or 1 (one)**

14(a) Derivation. (6)

(b) The equation of the vibrating string is

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2} \quad (1)$$

Writing boundary conditions and initial conditions. (2)

The solution is of the form

$$Y(x,t) = (c_1 \cos px + c_2 \sin px) (c_3 \cos cpt + c_4 \sin cpt). \quad (1)$$

Applying the conditions and simplifying



$$y(x,t) = \frac{3v_0 l}{4\pi c} \sin\left(\frac{\pi x}{l}\right) \sin\left(\frac{\pi ct}{l}\right) - \frac{v_0 l}{12\pi c} \sin\left(\frac{3\pi x}{l}\right) \sin\left(\frac{3\pi ct}{l}\right) \quad (4)$$

Module 3

15(a) $U = \sin x \cosh y$. (7)

$$u_x = \cos x \cosh y \quad u_{xx} = -\sin x \cosh y$$

$$u_y = \sin x \sinh y \quad u_{yy} = \sin x \cosh y$$

$$u_{xx} + u_{yy} = 0 \quad U \text{ is harmonic.} \quad (2)$$

To find

$$V = \sin x \cosh y + c \quad (2)$$

$$\text{To find } f(z) = \sin z + ic \quad (3)$$

(b) $W = Z^2$

$$u = x^2 - y^2, \quad v = 2xy \quad (1)$$

$$\text{when } x = \frac{1}{2} \quad v^2 = \frac{1}{4} - u, \text{ which is a parabola.} \quad (2)$$

$$\text{When } x = 1 \quad v^2 = 4(1 - u), \text{ which is a parabola} \quad (2)$$

Hence the image of the strip $\frac{1}{2} \leq x \leq 1$ is the region between the parabolas

$$v^2 = \frac{1}{4} - u \text{ and } v^2 = 4(1 - u). \quad (2)$$

16(a) $W = \log z$ (8)

$$= u + iv = \frac{1}{2} \log(x^2 + y^2) + i \tan^{-1}\left(\frac{y}{x}\right) \quad (2)$$

$$\text{Finding } u_x, u_y, v_x, v_y \quad (2)$$

$$u_x = v_y \quad \text{and} \quad u_y = -v_x \quad (2)$$

CR equations are and partial derivatives are continuous. Hence

$$F(z) \text{ is analytic} \quad (2)$$

(b) $W = \frac{1}{z}$

$$x = \frac{u}{u^2 + v^2} \quad \text{and} \quad y = \frac{-v}{u^2 + v^2} \quad (2)$$

$$x^2 + y^2 - 6x = 0 \quad (6)$$

$$\left(\frac{u}{u^2 + v^2}\right)^2 + \left(\frac{-v}{u^2 + v^2}\right)^2 - 6u = 0 \quad (2)$$



After simplification

$$u = \frac{1}{6}, \text{ which is a straight line in the } W \text{ plane.} \quad (2)$$

Module 4

$$17(a) \oint_C \frac{f(z)dz}{(z-z_0)^4} dz = \frac{2\pi i}{3!} f^3(z_0) \quad (1) \quad (7)$$

$$\oint_C \frac{\sin 2z}{(z-0)^4} dz = \frac{2\pi i}{3!} f^3(0) \quad (2)$$

Where $f(z) = \sin 2z$.

Finding $f^3(0)$ and substituting in the formula (3)

$$\oint_C \frac{\sin 2z dz}{(z-0)^4} = \frac{-8}{3} \pi i \quad (1)$$

$$(b) f(z) = \frac{1}{1+z} = \frac{1}{1+z+i-i} \quad (1) \quad (7)$$

$$= \frac{1}{1+i} \left(1 + \left(\frac{z-i}{1+i}\right)\right)^{-1} \quad (3)$$

$$= \frac{1}{1+i} - \frac{(z-i)}{(1+i)^2} + \frac{(z-i)^2}{(1+i)^3} + \dots \quad (3)$$

$$18(a) \int_0^{1+i} (x - y + ix^2) dz \quad (7)$$

$$y = x^2$$

$$dy = 2x dx$$

$$z = x + iy$$

$$dz = dx + i dy$$

$$dz = dx + i 2x dx = (1 + 2ix) dx \quad (2)$$

$$\int_0^{1+i} (x - y + ix^2) (1 + 2ix) dx = \int_0^{1+i} (x - x^2 + ix^2) (1 + 2ix) dx$$

(3)

$$= \frac{-1}{3} + \frac{i}{2} \quad (2)$$

$$(b) \quad (7)$$

The function $\log z$ is analytic at all points inside and on the circle

$$|z - 3| = 2. \quad (1).$$



The singularity $z = 4$ lie inside the circle $|z - 3| = 2$. (1)

Applying Cauchy's integral formula (4)

$$\oint_C \frac{\log z}{(z-4)^2} = \frac{\pi i}{2} \quad (1)$$

Module 5

$$19(a) \quad f(z) = \frac{z^2 - 1}{z^2 + 5z + 6} = 1 - \left(\frac{-3}{z+2} + \frac{8}{z+3} \right) \quad (1) \quad (5)$$

$$= 1 - \left(\frac{-3}{z} \left(1 + \frac{2}{z}\right)^{-1} + \frac{8}{3} \left(1 + \frac{z}{3}\right)^{-1} \right). \quad (2)$$

$$= 1 - \left(\frac{8}{3} - \frac{8z}{9} + \frac{8z^2}{27} + \dots + \frac{-3}{z} + \frac{6}{z^2} - \frac{12}{z^3} \dots \right) \quad (2)$$

$$(b) \quad \int_0^{2\pi} \frac{d\theta}{\sqrt{2} - \cos \theta} \quad (1) \quad (9)$$

Consider the contour $C: |z| = 1$.

$$Z = e^{i\theta} \quad d\theta = \frac{dz}{iz}$$

$$\cos \theta = \frac{z^2 + 1}{2z}$$

$$I = \int_0^{2\pi} \frac{d\theta}{\sqrt{2} - \cos \theta} = 2i \int_{|z|=1} \frac{dz}{z^2 - 2\sqrt{2}z + 1} \quad (2).$$

$$\text{Poles } z = \sqrt{2} + 1, \sqrt{2} - 1. \quad (1).$$

$$\sqrt{2} - 1 \text{ lie inside the region} \quad (1)$$

$$\text{Residue } \frac{-1}{2}. \quad (2).$$

Applying Residue theorem and finding the answer

$$I = 2\pi. \quad (2)$$

$$20(a) \quad \oint_C \frac{z-23}{z^2-4z-5} dz \text{ where } C: |z-2-i| = 3.2. \quad (5)$$

$$\text{Singular points } z = 5, -1. \quad (1)$$

$$\text{Residue of } f(z) \text{ at } z = 5 \text{ is } -3. \quad (1)$$

$$\text{Residue of } f(z) \text{ at } z = -1 \text{ is } 4. \quad (1)$$

$$\oint_C \frac{z-23}{z^2-4z-5} dz = 2\pi i. \quad (2)$$



(b)

$$\int_0^{\infty} \frac{(x^2 + 2)dx}{(x^2 + 1)(x^2 + 4)} = \frac{1}{2} \int_{-\infty}^{\infty} \frac{(x^2 + 2)dx}{(x^2 + 1)(x^2 + 4)} \quad (1)$$

(9)

Consider the contour integral

$I = \int_C f(z)dz$ where $f(z) = \frac{(z^2 + 2)dz}{(z^2 + 1)(z^2 + 4)}$. where C is the path consisting of the semi circular path C_R along with the diameter from $-R$ to $+R$.

(1).

Poles are $z = \pm i$ and $z = \pm 2i$.

(1)

 $z = i$ and $z = 2i$ lie inside the region.

(1)

Residue of $f(z)$ at $z = i$ is $\frac{1}{6i}$

(1)

Residue of $f(z)$ at $z = 2i$ is $\frac{1}{6i}$.

(1)

Applying Residue theorem and finding the value of the integral

$$\text{as } \int_0^{\infty} \frac{(x^2 + 2)dx}{(x^2 + 1)(x^2 + 4)} = \frac{\pi}{3} \quad (3).$$

NB: Any alternative correct answers will get full marks.